

WILLIAMSPORT, PA, USA

WMO: 725140

Lat: 41.243N

Lon: 76.922W

Elev: 159

StdP: 99.44

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14778

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-12.2	-21.6	0.6	-12.9	-19.0	0.7	-10.3	11.0	-0.3	9.9	-0.6	3.0	270	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.6	22.4	31.0	21.8	29.4	21.1	24.2	29.7	23.4	28.4	22.6	27.2	3.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.6	26.7	21.8	16.8	25.9	21.1	16.1	25.1	73.5	29.9	70.3	28.4	67.3	27.1	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.2	7.2	-18.3	35.4	3.7	1.7	-21.0	36.7	-23.1	37.7	-25.2	38.6	-27.9	39.9		
(5)			-18.9	25.8	3.5	0.8	-21.4	26.3	-23.5	26.8	-25.4	27.3	-28.0	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-2.6	-1.0	3.8	10.3	16.0	20.8	23.1	22.2	18.3	11.8	5.6	0.5		
	DBStd	9.97	5.61	5.10	5.27	4.70	4.31	3.36	2.82	2.78	3.71	4.31	4.48	4.66		
	HDD10.0	1431	392	310	204	51	3	0	0	0	29	146	296			
	HDD18.3	3194	649	543	451	243	101	15	1	3	46	207	383	552		
	CDD10.0	1722	1	1	11	61	189	324	407	378	250	86	13	2		
	CDD18.3	443	0	0	0	3	28	89	149	123	45	6	0	0		
	CDH23.3	3676	0	0	8	74	297	764	1274	918	310	31	0	0		
	CDH26.7	1194	0	0	2	20	92	238	464	289	84	4	0	0		
(14) Wind	WSAvg	2.7	3.2	3.2	3.4	3.3	2.7	2.4	2.1	1.9	2.0	2.4	2.7	3.0		
(15) Precipitation	PrecAvg	1040	69	64	82	85	97	102	104	88	101	82	93	73		
	PrecMax	1504	210	214	151	192	185	408	245	195	320	244	197	179		
	PrecMin	764	13	14	22	18	20	16	30	24	12	5	21	17		
	PrecStd	178	43	41	32	45	44	62	46	41	68	52	40	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	16.3	23.4	29.3	31.9	33.5	35.2	34.3	32.0	26.7	20.2	16.8		
		MCWB	13.5	10.1	14.1	17.5	20.8	23.2	23.8	23.2	22.0	19.8	14.7	14.1		
	2%	DB	10.2	10.9	18.5	25.3	29.3	31.3	32.8	31.6	29.1	23.4	17.5	11.9		
		MCWB	7.6	6.3	11.2	15.3	19.6	21.9	22.8	22.1	21.0	18.1	13.4	9.5		
	5%	DB	7.0	8.2	15.0	21.9	26.8	29.4	31.0	29.6	26.8	21.3	15.1	9.3		
		MCWB	4.8	4.6	9.7	13.5	18.4	21.1	22.1	21.4	19.9	16.5	11.2	7.1		
	10%	DB	4.8	5.9	12.1	19.1	24.1	27.7	29.3	28.0	24.7	19.0	12.9	7.1		
		MCWB	2.9	3.2	7.8	11.6	17.1	20.2	21.4	20.8	18.9	14.9	9.6	4.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.4	11.6	15.7	19.1	22.7	24.6	25.6	25.2	23.9	21.4	16.5	14.3		
		MCDB	14.9	14.8	21.7	26.4	29.2	30.4	32.1	31.1	28.9	25.0	18.8	16.3		
	2%	WB	8.0	7.3	12.8	16.8	21.0	23.5	24.4	23.9	22.5	19.5	14.3	10.3		
		MCDB	9.3	9.8	17.0	22.6	26.7	28.8	30.2	28.8	26.7	22.4	16.5	11.6		
	5%	WB	5.1	5.2	10.3	15.0	19.8	22.5	23.5	22.9	21.4	17.6	12.1	7.5		
		MCDB	6.8	7.8	14.2	20.1	25.0	27.4	28.7	27.3	24.8	20.4	14.3	9.1		
	10%	WB	3.0	3.3	7.9	13.0	18.4	21.5	22.7	22.1	20.3	15.5	10.1	5.2		
		MCDB	4.6	5.7	11.7	17.8	22.6	26.0	27.2	26.0	23.3	18.3	12.6	7.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.9	10.1	12.1	12.0	11.4	11.5	11.0	11.0	10.6	9.0	7.1		
		MCDBR	10.1	12.3	15.1	17.0	15.6	13.6	13.2	12.9	13.3	13.4	12.5	9.6		
	5% WB	MCWBR	8.4	8.7	9.1	9.0	7.4	6.0	5.2	5.4	6.3	7.7	8.5	7.9		
		MCWBR	9.6	10.7	13.6	14.4	13.1	11.5	11.2	10.7	10.9	11.0	10.4	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.316	0.329	0.364	0.388	0.422	0.478	0.486	0.477	0.390	0.372	0.329	0.309		
	taud		2.372	2.325	2.318	2.302	2.236	2.112	2.105	2.113	2.340	2.390	2.474	2.450		
	Ebn at Noon		830	875	882	883	857	807	795	789	844	817	814	812		
	Edh at Noon		83	101	115	126	137	156	156	150	112	95	76	72		
(44) All-Sky Solar Radiation	RadAvg		1.65	2.53	3.51	4.53	5.16	5.61	5.72	5.03	4.09	2.76	1.88	1.35		
	RadStd		0.13	0.29	0.30	0.49	0.55	0.49	0.37	0.31	0.52	0.32	0.23	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+123	N/A
(47)	Regional (38 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+110	N/A

Nomenclature: See separate page